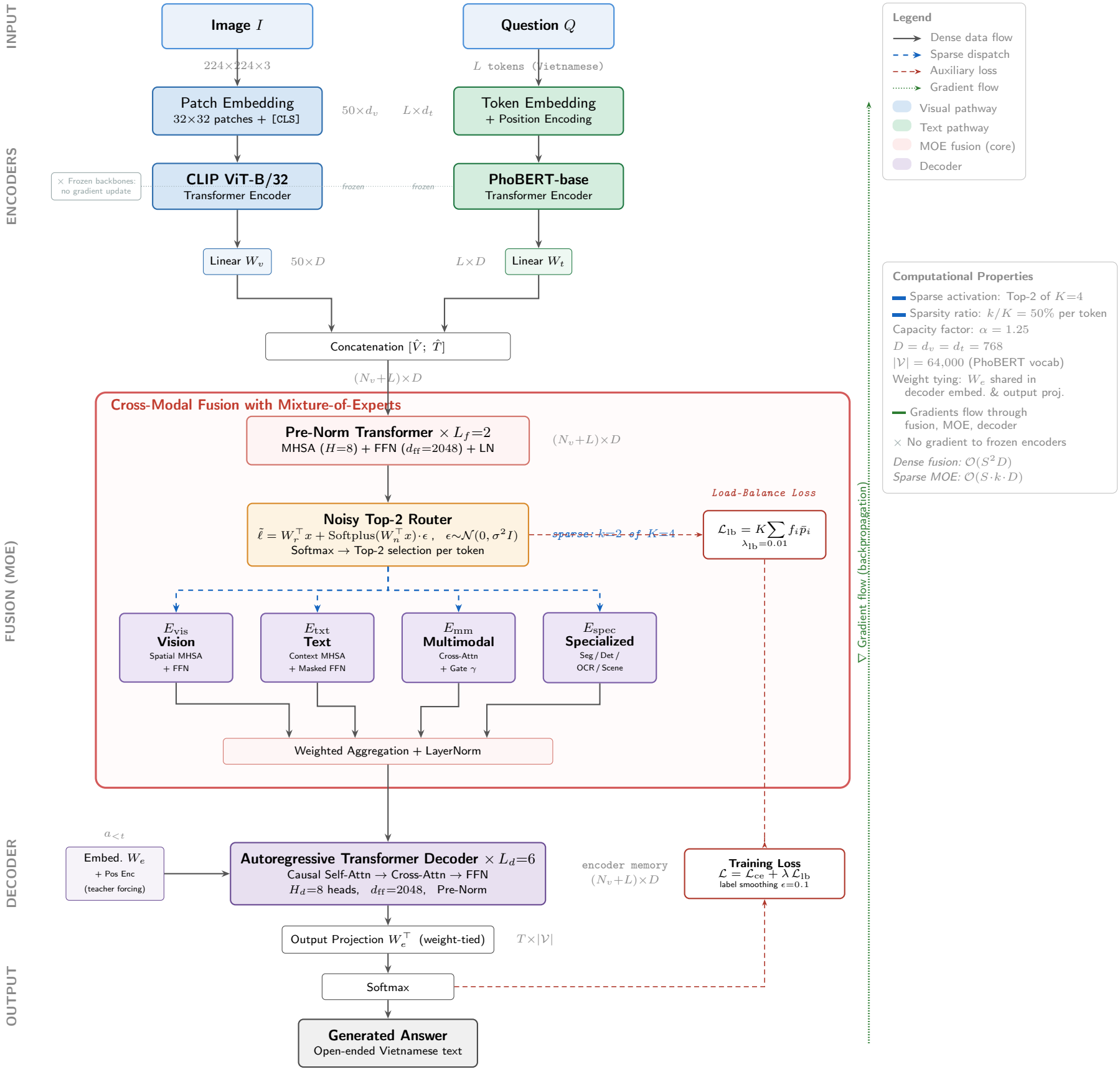
INPUT
ENCODERS
FUSION (MOE)
DECODER
OUTPUT
Image I
Question Q
$224\times224\times3$
L tokens (Vietnamese)
Patch Embedding
32×32 patches + [CLS]
$50\times d_v$
$L\times d_t$
Token Embedding
+ Position Encoding
× Frozen backbones: no gradient update
CLIP ViT-B/32
Transformer Encoder
frozen
frozen
PhoBERT-base
Transformer Encoder
Linear W_v
$50\times D$
$L\times D$
Linear W_t
Concatenation $[\hat{V};\ \hat{T}]$
$(N_v+L)\times D$
Cross-Modal Fusion with Mixture-of-Experts
Pre-Norm Transformer $\times L_f=2$
MHSA (H=8) + FFN (d_{ff}=2048) + LN
$(N_v+L)\times D$
Noisy Top-2 Router
$\tilde{\ell} = W_r^\top x + \text{Softplus}(W_n^\top x)\cdot\epsilon\ ,\quad \epsilon\sim\mathcal{N}(0,\sigma^2 I)$
Softmax → Top-2 selection per token
sparse: k=2 of K=4
Load-Balance Loss
$\mathcal{L}_{\text{lb}} = K\sum f_i\bar{p}_i$
$\lambda_{\text{lb}}=0.01$
E_{vis} Vision
Spatial MHSA + FFN
E_{txt} Text
Context MHSA + Masked FFN
E_{mm} Multimodal
Cross-Attn + Gate γ
E_{spec} Specialized
Seg / Det / OCR / Scene
Weighted Aggregation + LayerNorm
$a_{<t}$
Embed. W_e
+ Pos Enc
(teacher forcing)
Autoregressive Transformer Decoder $\times L_d=6$
Causal Self-Attn → Cross-Attn → FFN
H_d=8 heads, d_{ff}=2048, Pre-Norm
encoder memory
$(N_v+L)\times D$
Training Loss
$\mathcal{L} = \mathcal{L}_{\text{ce}} + \lambda\,\mathcal{L}_{\text{lb}}$
label smoothing ϵ=0.1
Output Projection $W_e^\top$ (weight-tied)
$T\times|\mathcal{V}|$
Softmax
Generated Answer
Open-ended Vietnamese text
∇ Gradient flow (backpropagation)
Legend
Dense data flow
Sparse dispatch
Auxiliary loss
Gradient flow
Visual pathway
Text pathway
MOE fusion (core)
Decoder
Computational Properties
Sparse activation: Top-2 of K=4
Sparsity ratio: $k/K = 50\%$ per token
Capacity factor: $\alpha = 1.25$
$D = d_v = d_t = 768$
$|\mathcal{V}| = 64{,}000$ (PhoBERT vocab)
Weight tying: W_e shared in decoder embed. & output proj.
Gradients flow through fusion, MOE, decoder
× No gradient to frozen encoders
Dense fusion: $\mathcal{O}(S^2 D)$
Sparse MOE: $\mathcal{O}(S\cdot k\cdot D)$